

Copied Inertial Observations

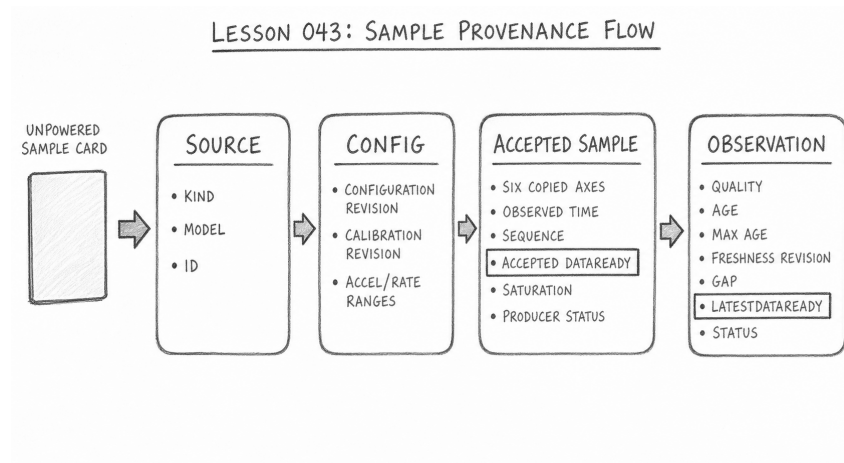
Preserve the evidence before interpreting motion

E0 SYNTHETIC REPLAY — NO POWERED INERTIAL MODULE CLAIM

Lesson	043 — component	Time	90–120 minutes
Prerequisites	fixed-width values, status, time	Board	compile-only Mega replay
Evidence	named memory result cells	Boundary	no bus, pins, wiring, or sensor

Responsibility. Copy and validate one six-axis sample. Retain its source, units, range, revisions, sequence, timestamp, saturation, producer status, and derived freshness. Do not rotate, filter, calibrate, integrate, or infer a physical device.

1 Read one complete observation



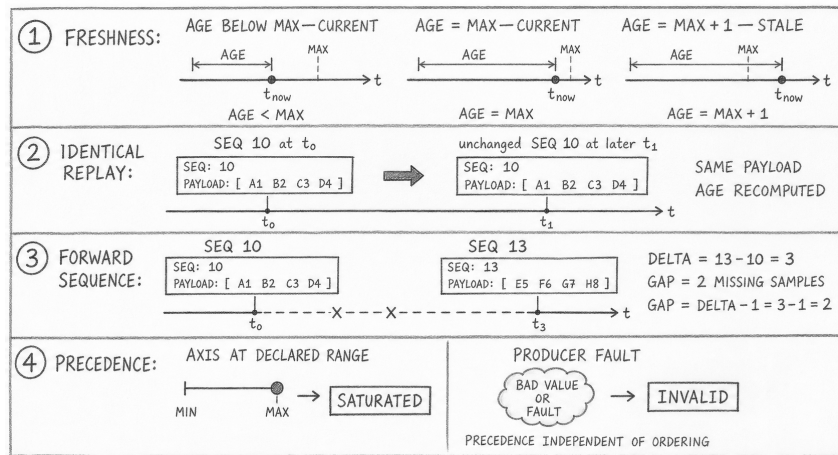
Alternative text: a pencil drawing follows a synthetic source identity into one copied acceleration and angular-rate sample, then into derived quality, age, and sequence-gap evidence without contacting hardware.

The three acceleration fields use micro-g. The three angular-rate fields use millidegrees per second. A vector gains its unit from the named field that contains it; it is not a unit-free math type. `observedAt` and `age` use milliseconds.

Evidence	Meaning
source domain	kind, model, source ID, ranges, configuration and calibration revisions
accepted payload	timestamp, sequence, <code>sample.dataReady</code> , axes, saturation, and producer status
freshness contract	maximum age and its nonzero revision
latest evidence	<code>latestDataReady</code> , quality, age, gap, and status

Only the `SyntheticFixture/Synthetic` pair is accepted in E0. The MPU6050 and QMI8658 names are negative-test seams, not supported adapters.

2 Predict the quality timeline



Alternative text: a pencil timeline shows an accepted synthetic sample, an identical same-sequence replay aging from current to stale, a separate saturated sample, and a producer fault; rejected evidence leaves the last accepted sample unchanged.

Condition	Predicted result
age less than or equal to maximum	Current , status OK
age greater than maximum	Stale , status OK
same-sequence not-ready poll	retained ready payload; latest not ready; Stale
axis exactly at its declared range	Saturated , status OK
non-OK producer status	Invalid , producer failure retained
malformed identity, time, or value	Invalid , invalid argument

Precedence matters. Structural invalidity wins over staleness. Producer failure makes numeric payload untrustworthy. For structurally valid payload, saturation wins over age because bounded measurement meaning is already lost.

3 Replay worksheet

Use the constant frames in the Lesson 043 replay sketch. Before reading the result cells, complete the prediction columns. Units belong in every numeric answer.

Frame	Prediction	Observed cell	Interpretation
1	quality: _____	_____	_____
2	gap: _____	_____	_____
3	ready/latest: _____	_____	_____
4	axes: _____	_____	_____
5	saturation: _____	_____	_____
6	age: _____ms	_____	_____
7	status: _____	_____	_____

The result cells are memory evidence for a host harness or simulator. The sketch is compiled for packaging but is not run on a powered Mega in E0. Serial output, a physical LED, and a debugger are not required or claimed.

4 Work the invariants

Freshness calculation

For wrap-safe forward time, calculate $age = policyTime - observedAt$.

policyTime (ms)	observedAt (ms)	maximumAge (ms)	quality
_____	_____	_____	_____
_____	_____	_____	_____

At exactly **maximumAge**, predict **Current**. One millisecond later, predict **Stale**. An exact semantic replay may make this transition without creating a new sample or sequence gap.

Sequence calculation

For a forward modular sequence delta d , the recorded gap is $d - 1$. Delta zero is accepted only when every named sample field is semantically equal. The sole changed-field exception is a same-domain, same-sequence **dataReady** true-to-false poll: it retains the accepted ready payload, sets **latestDataReady** false, and publishes stale evidence. Delta 0x80000000 is ambiguous and invalid. The poll creates no new sample, sequence, or gap.

previous sequence	candidate sequence	predicted gap	accept/reject
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____

5 Diagnose without inventing evidence

1. Change one axis but retain the same sequence. Predict rejection. Verify that the previously accepted snapshot and history remain unchanged.
2. Advance policy time while replaying the exact sample. Predict a later age and no new gap. Continue until quality becomes **Stale**.
3. Change only **dataReady** from true to false at the same sequence. Predict retained axes and accepted readiness true, but latest readiness false and quality **Stale**.
4. Place one axis exactly at its declared range and mark matching saturation. Predict **Saturated**; then remove the marker and predict structural rejection.
5. Set **sourceId**, a revision, or a declared range to zero. Predict **Invalid**. Do not describe the result as a sensor reading.
6. Supply **HardwareFailure**. Predict that producer failure dominates the numeric payload and that accepted history is preserved.

Safe-state result. This lesson acquires no hardware resource, so reset clears policy history and returns the canonical invalid snapshot. It cannot prove electrical shutdown, pin state, or power removal.

6 Separate acquisition from safe state

Stage	Predict	Observe and interpret
Acquire	no endpoint; initialize valid config	status/result cells: _____
Observe	copy one bounded synthetic frame	quality/age/gap: _____
Reset	history clears; configuration remains	canonical snapshot: _____

This separation is deliberate. Software initialization is resource-acquisition evidence for a pure object. Reset is its software safe state. Neither is a substitute for a future adapter's bus rollback or a module's electrical inactive state.

7 Malformed and deferred cases

Case	Diagnosis
stale	valid copied value exceeded its declared freshness window
not ready	same-sequence poll retains accepted ready payload; latest readiness is false; no new sequence or gap
saturated	valid copied axis reached its declared range; magnitude is bounded, not precise
unsteady	not classified here; Lesson 044 owns stationary-orientation eligibility
producer fault	source status is non-OK; numeric axes are not trusted
MPU/QMI source	unsupported in E0; exact specimen and adapter qualification remain open

8 Blank E1 acceptance record

Do not prefill from compilation, replay, a datasheet, or this PDF.

Required field	Bench record
board revision / exact specimen	_____
module and chip markings	_____
supply / logic levels / instrument	_____
tool version / tested commit	_____
resource-acquisition prediction	_____
resource-acquisition observation	_____
safe-state prediction	_____
safe-state observation	_____
interpretation / reviewer / date	_____

9 Evidence links

- Copied-observation public interface
- Deterministic Lesson 043 tests
- Compile-only synthetic replay
- Lessons 043–045 design and open gates

The searchable HTML lesson is the hardware-independent accessible route. This untagged print edition makes no PDF/UA, powered-module, or physical acceptance claim.